

SANLIURFA LIGHT RAIL SYSTEM - KARAKOPRU / HALILIYE-EYYUBIYE LINE, PHASE 1

Signalling and Vehicle Tracking System - Quotation - POINT LINK

Quotation addressed to	OZVER INSAAT A.S.
Consultant	MEP Center - Serhat CELIK
Employer	Sanliurfa Metropolitan Municipality
Tender no.	2026/1327584 (13 October 2026)
Quotation date	1 October 2026
Submission deadline	7 October 2026
Validity	180 calendar days
Delivery	DAP Sanliurfa site
Currency	EUR (VAT excluded)

DOCUMENTS USED TO PREPARE THIS QUOTATION

- 1
- 2
- 3
- 4
- 5
- BOQ URF_1Etap_SINYALIZASYON_Teklif_Talebi.xlsx (29.09.2026), items 228-235 (Seq. No 228 to 235).
- Technical Specification URF-000-000-00-94-103 Section 3 - Signalling and URF-000-000-00-94-104 Section 4 - Control & Communication.
- Tender drawings and schematic sheets from the shared drive: lineside schematics, route schematics, depot layout, transformer/equipment location drawings.
- Your enquiry e-mail of 1 October 2026 (21:07), including the note that quantities are not binding.
- Two previous CASCO quotations for comparable tram schemes: **Konya Alaaddin** (EUR 4,440,670, 10.06.2026) and **Adapazari-Serdivan** (EUR 3,364,100, 13.06.2026).

GRAND TOTAL SUMMARY

MATERIAL	EUR 4,410,023	CNY 33,380,346 / TRY 244,107,121
LABOUR (installation & services)	EUR 2,518,959	installation EUR 812,459 + design, testing, commissioning, training EUR 1,706,500
FREIGHT & INSURANCE (DAP)	EUR 352,802	provisional, 8% of material value
GRAND TOTAL	EUR 7,281,784	CNY 55,117,276 excl. VAT / CNY 62,282,522 incl. VAT
PER VEHICLE (15 vehicles)	EUR 82,019	material EUR 56,403 + labour EUR 25,616

Basis: DAP Sanliurfa site, EUR, VAT excluded. Material and labour are shown separately as required by BOQ note 1. The "Installed Unit EUR" column = (material + installation + commissioning services) / quantity, i.e. the installed unit price requested in your e-mail. **DAP scope:** we cover export clearance, sea and inland freight, insurance and delivery to the Sanliurfa site; import clearance, customs duty and import taxes are for the buyer's account. **FX:** EUR/CNY 7.5692, EUR/TRY 55.3528 (Sina Finance, 1 October 2026, 21:49 GMT+8). CNY and TRY amounts are converted for reference only.

SUMMARY BY ITEM

Seq.	Item	Description	Material EUR	Labour EUR	Total EUR	CNY excl. VAT	TRY
228	3001	Signalling System Design	0	540,000	540,000	4,087,368	29,890,512
229	3001.A	Lineside Signalling System	1,799,864	799,152	2,599,016	19,672,473	143,862,824
230	3001.B	Depot Area and Facilities Signalling System	671,108	301,262	972,370	7,360,067	53,823,430
231	3001.C	On-Board Signalling System	567,000	281,725	848,725	6,424,169	46,979,305
232	3001.D	Signalling Central Equipment & System Software	700,800	226,805	927,605	7,021,228	51,345,534
233	3002	Vehicle Tracking System (AVLS) - On-Board	279,050	102,515	381,565	2,888,142	21,120,691
234	3003	Tram Vehicle Tracking System - Central Equipment and Software	392,200	117,500	509,700	3,858,021	28,213,322
235	3004	Vehicle Tracking System Design	0	150,000	150,000	1,135,380	8,302,920
		Subtotal (items)	4,410,023	2,518,959	6,928,982	52,446,848	383,538,538
		Freight & insurance (DAP)	-	352,802	352,802	2,670,428	19,528,570
		GRAND TOTAL	4,410,023	2,871,761	7,281,784	55,117,276	403,067,108

PER-VEHICLE UNIT PRICE (e-mail note 5)

Item	Material EUR	Labour EUR	Total EUR	Installed unit EUR per vehicle
3001.C On-board signalling (15 vehicles)	567,000	281,725	848,725	56,582
3002 AVLS on-board (15 vehicles)	279,050	102,515	381,565	25,438
Total (15 vehicles)	846,050	384,240	1,230,290	82,019

BILL OF QUANTITIES - DETAIL

3001 - Signalling System Design (Seq. No 228)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Operations (signalling) simulation – headway and line plan verification (4/3/2.5 min)	set	1	-	-	25,000.00	25,000	25,000	25,000.00	EUR	-	
System architecture, route/interlocking tables (main line 6 routes + depot 15 routes), point zone and junction function definitions	set	1	-	-	60,000.00	60,000	60,000	60,000.00	EUR	-	
Detail design: 3 point zones + 5 junctions + depot entrance + 3 pedestrian crossings + depot layout, loop layout calculations	pafta-set	1	-	-	55,000.00	55,000	55,000	55,000.00	EUR	-	
Cable drawings, cable schedules, cabinet internal schematics, earthing/isolation transformer concept, EMC (EN 50121-4)	set	1	-	-	70,000.00	70,000	70,000	70,000.00	EUR	-	
RAMS / hazard log / safety case – main line SIL3, depot min SIL2 (EN 50126/50128/50129)	set	1	-	-	90,000.00	90,000	90,000	90,000.00	EUR	-	
Independent safety assessment (ISA) and certification – engineer-approved body	set	1	-	-	180,000.00	180,000	180,000	180,000.00	EUR	-	
Test procedures (FAT/SAT/integration), O&M documentation, as-built drawings	set	1	-	-	45,000.00	45,000	45,000	45,000.00	EUR	-	
KET regulation (short-range radio – loop/transponder) compliance / permit file	set	1	-	-	15,000.00	15,000	15,000	15,000.00	EUR	-	
			3001 Subtotal	0		540,000	540,000				

3001.A - Lineside Signalling System (Seq. No 229)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Electric point machine (IP67, internal mechanical lock, >=15 km/h facing approach, hand-crank operable, steel housing, three-position detection) – R50 depot connection	takim	4	6,625.00	26,500	1,325.00	5,300	31,800	7,950.00	EUR	CASCO tram-type (model TBC)	
Electric point machine (IP67, internal mechanical lock, three-position detection) – R100 (C5 crossover + M17-M20)	takim	8	6,625.00	53,000	1,325.00	10,600	63,600	7,950.00	EUR	CASCO tram-type (model TBC)	
Point machine connecting/drive rods + point lock (clamp lock) + detection rod set	takim	12	1,850.00	22,200	370.00	4,440	26,640	2,220.00	EUR	TBD (to be confirmed)	
Manual point operating mechanism (lever, key lock, position contact) – main line	takim	4	2,400.00	9,600	480.00	1,920	11,520	2,880.00	EUR	TBD (to be confirmed)	
Manual point position monitoring box (IO/RTU – position change reporting to TCC/SCADA)	adet	1	6,500.00	6,500	975.00	975	7,475	7,475.00	EUR	TBD (to be confirmed)	
Local control and command cabinet SKP (GRP IP54, special lock, SIL3 microprocessor point control unit, I/O, point driver, isolation transformer, DC power, surge arrester)	adet	3	145,000.00	435,000	21,750.00	65,250	500,250	166,750.00	EUR	CASCO SIL3/SIL2 platform (model TBC)	
Interrogator (transmitter/receiver) unit – inside SKP	adet	4	9,500.00	38,000	1,425.00	5,700	43,700	10,925.00	EUR	TBD (to be confirmed)	
Receiving inductive loop (route request) – point zones	adet	11	2,400.00	26,400	480.00	5,280	31,680	2,880.00	EUR	TBD (to be confirmed)	Loop quantity estimated (Ongoru); to be confirmed against detail design.
Track circuit / mass-weight detector (point locking, zone occupancy/clearance)	adet	34	4,450.00	151,300	890.00	30,260	181,560	5,340.00	EUR	CASCO AxC (model TBC)	Axle counters offered in lieu of track circuits; re-quote if track circuits / mass detectors are mandated.
Main line point signal head 3-aspect LED (Stop / Right / Left) + Request-received / Occupied lamp	takim	11	1,850.00	20,350	462.50	5,088	25,438	2,312.50	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Signal post (corrosion resistant, maintenance free) – point zones	adet	11	650.00	7,150	162.50	1,788	8,938	812.50	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Local request (manual) switch – special key lock, mounted on signal post – point zones	adet	11	480.00	5,280	120.00	1,320	6,600	600.00	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Road junction signalling control cabinet TSP (GRP IP54, interrogator, loop cards, TSKP relay interface, isolation transformer, DC power, network port)	adet	5	68,000.00	340,000	10,200.00	51,000	391,000	78,200.00	EUR	TBD (to be confirmed)	
Receiving inductive loop (ON/OFF priority request) – junctions	adet	24	2,000.00	48,000	400.00	9,600	57,600	2,400.00	EUR	TBD (to be confirmed)	Loop quantity estimated (Ongoru); to be confirmed against detail design.
Tram signal head 2-aspect LED (Stop-Go) – junctions	adet	12	1,450.00	17,400	362.50	4,350	21,750	1,812.50	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Tram signal post – junctions	adet	12	650.00	7,800	162.50	1,950	9,750	812.50	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Local command switch (priority request, special key lock) – junctions	adet	12	480.00	5,760	120.00	1,440	7,200	600.00	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Pedestrian crossing tram warning set (single red flashing beacon + 'tram approaching' sign plate) – stops not adjacent to a junction	adet	6	1,250.00	7,500	312.50	1,875	9,375	1,562.50	EUR	TBD (to be confirmed)	Quantity estimated (Ongoru); to be confirmed against detail design.
Pedestrian crossing warning control box + activation loop (2 loops/crossing)	takim	3	4,800.00	14,400	720.00	2,160	16,560	5,520.00	EUR	TBD (to be confirmed)	Loop quantity estimated (Ongoru); to be confirmed against detail design.
TSKP road junction controller (with tram priority phase) – new or replacement of existing	adet	6	22,000.00	132,000	3,300.00	19,800	151,800	25,300.00	EUR	Local / third party (TBD)	Road traffic equipment - provisional; local existing type to be verified.
Road traffic signal head 300 type, overhead (with mast arm)	adet	20	1,850.00	37,000	462.50	9,250	46,250	2,312.50	EUR	Local / third party (TBD)	Road traffic equipment - provisional; local existing type to be verified.
Road traffic signal head 200 type (plain + arrow), pole mounted	adet	44	720.00	31,680	180.00	7,920	39,600	900.00	EUR	Local / third party (TBD)	Road traffic equipment - provisional; local existing type to be verified.
Pedestrian signal head + pedestrian push button	takim	44	380.00	16,720	95.00	4,180	20,900	475.00	EUR	Local / third party (TBD)	Road traffic equipment - provisional; local existing type to be verified.
Traffic signal pole 3.5 m (straight) / 6 m mast arm (L)	adet	64	620.00	39,680	155.00	9,920	49,600	775.00	EUR	Local / third party (TBD)	Road traffic equipment - provisional; local existing type to be verified.
Road traffic signal cables (TSKP – signal heads, multi-core)	m	2,695	14.00	37,730	8.40	22,638	60,368	22.40	EUR	TBD (to be confirmed)	Road cables/ducts; quantity includes 10% waste and installation allowance.
Road signal cable duct HDPE D63 (junction crossings)	m	1,210	7.00	8,470	4.20	5,082	13,552	11.20	EUR	TBD (to be confirmed)	Road cables/ducts; quantity includes 10% waste and installation allowance.
Loop feeder cable (screened twisted pair 2x2x1.0 – between loop and TSP/SKP)	m	5,468	6.50	35,542	3.90	21,325	56,867	10.40	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Loop cable (loop conductor in track / trackbed + protection channel)	adet	41	1,100.00	45,100	660.00	27,060	72,160	1,760.00	EUR	TBD (to be confirmed)	Loop quantity estimated (Ongoru); to be confirmed against detail design.
Track circuit / detector cable (screened twisted pair 2x2x1.0)	m	2,673	6.50	17,374	3.90	10,425	27,799	10.40	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Point machine power + control cable (e.g. 12x2.5 signalling cable)	m	541	19.00	10,279	11.40	6,167	16,446	30.40	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Signal head cable (12x1.5 signalling cable)	m	1,348	13.00	17,524	7.80	10,514	28,038	20.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Local command switch cable (7x1.5)	m	1,348	8.50	11,458	5.10	6,875	18,333	13.60	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
TSP/SKP – TSKP interface cable (volt-free relay contacts 19x1.5)	m	198	17.00	3,366	10.20	2,020	5,386	27.20	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Manual point position contact cable (7x1.5)	m	117	8.50	994	5.10	597	1,591	13.60	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Pedestrian crossing flashing beacon cable (3x1.5)	m	264	4.50	1,188	2.70	713	1,901	7.20	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cabinet supply cable NYY 3x6 (from substation UPS panel to SKP/TSP/IO boxes)	m	5,905	11.00	64,955	6.60	38,973	103,928	17.60	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Cable protection duct HDPE D50 (equipment – lineside cable route connections, system specific)	m	2,327	5.50	12,798	3.30	7,679	20,478	8.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cable protection duct HDPE D110 (under track / under road crossings, system specific)	m	605	13.00	7,865	7.80	4,719	12,584	20.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cabinet and signal post foundations (small concrete base + anchorage)	adet	100	260.00	26,000	130.00	13,000	39,000	390.00	EUR	TBD (to be confirmed)	Small concrete base and anchorage only; excludes excavation and road reinstatement.
Signalling testing, commissioning, traffic signal coordination (each junction), integrated testing and 15-day trial operation – lineside	set	1	-	-	360,000.00	360,000	360,000	360,000.00	EUR	-	
			3001.A Subtotal	1,799,864		799,152	2,599,016				

3001.B - Depot Area and Facilities Signalling System (Seq. No 230)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Electric point machine – depot (IP67, hand-crank operable, three-position detection, depot type 15 km/h)	takim	20	5,900.00	118,000	1,180.00	23,600	141,600	7,080.00	EUR	CASCO tram-type (model TBC)	
Point machine drive/detection rod set + point lock – depot	takim	20	1,650.00	33,000	330.00	6,600	39,600	1,980.00	EUR	TBD (to be confirmed)	
Manual point operating mechanism (lever + lock + position indicator/contact) – depot	takim	20	2,200.00	44,000	440.00	8,800	52,800	2,640.00	EUR	TBD (to be confirmed)	
Depot signalling control unit SKP (SIL2 microprocessor, GRP IP54, I/O, point drivers, isolation transformer, DC power) + 1 manual point IO box	adet	2	110,000.00	220,000	16,500.00	33,000	253,000	126,500.00	EUR	CASCO SIL3/SIL2 platform (model TBC)	
Depot area manual route request panel MSP (schematic + push buttons, linked to SKP)	adet	4	9,500.00	38,000	1,425.00	5,700	43,700	10,925.00	EUR	TBD (to be confirmed)	
Depot signal head (point position + lock status indication, LED) + post	takim	20	1,600.00	32,000	400.00	8,000	40,000	2,000.00	EUR	TBD (to be confirmed)	
Track circuit / mass detector – depot (point lock + zone boundaries)	adet	24	4,450.00	106,800	890.00	21,360	128,160	5,340.00	EUR	CASCO AxC (model TBC)	Axle counters offered in lieu of track circuits; re-quote if track circuits / mass detectors are mandated.
Point machine power + control cable (12x2.5) – depot	m	552	19.00	10,488	11.40	6,293	16,781	30.40	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Manual point position contact cable (7x1.5) – depot	m	959	8.50	8,152	5.10	4,891	13,042	13.60	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Depot signal head cable (12x1.5)	m	992	13.00	12,896	7.80	7,738	20,634	20.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Track circuit / detector cable (screened twisted pair 2x2x1.0) – depot	m	816	6.50	5,304	3.90	3,182	8,486	10.40	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
MSP panel cable (24x1.5)	m	176	22.00	3,872	13.20	2,323	6,195	35.20	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
SKP supply cable NYY 3x6 (from depot substation UPS panel)	m	563	11.00	6,193	6.60	3,716	9,909	17.60	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Inter-SKP / SKP–workshop TCC data link (6-core multimode fibre optic)	m	662	9.00	5,958	1.35	894	6,852	10.35	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cable protection duct HDPE D110 (depot main signal cable route – 2 ducts)	m	880	13.00	11,440	7.80	6,864	18,304	20.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cable protection duct HDPE D50 (equipment connections + under track crossings)	m	1,452	5.50	7,986	3.30	4,792	12,778	8.80	EUR	TBD (to be confirmed)	Length includes 10% waste and installation allowance (BOQ note 7).
Cabinet / signal post / MSP foundations (small concrete base)	adet	27	260.00	7,020	130.00	3,510	10,530	390.00	EUR	TBD (to be confirmed)	Small concrete base and anchorage only; excludes excavation and road reinstatement.

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Depot route/interlocking table software (15+ routes), testing and commissioning – depot	set	1	-	-	150,000.00	150,000	150,000	150,000.00	EUR	-	
			3001.B Subtotal	671,108		301,262	972,370				

3001.C - On-Board Signalling System (Seq. No 231)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
On-board signalling operating panel (Right/Straight/Left/Ready-to-start buttons, route number selector, vehicle number switch) – 1 per cab	adet	30	3,200.00	96,000	480.00	14,400	110,400	3,680.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
On-board transceiver (transponder) + underframe antenna – 1 per vehicle end	adet	30	4,800.00	144,000	-	-	144,000	4,800.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
On-board signalling control/encoder unit (EN 50155)	adet	15	12,500.00	187,500	-	-	187,500	12,500.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Antenna cables, on-board wiring and mounting accessories (brackets, protection)	set	15	2,800.00	42,000	1,680.00	25,200	67,200	4,480.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Vehicle interface (speed/odometry, cab active, doors, driver/run number transfer to TCMS/AVLS)	set	15	6,500.00	97,500	975.00	14,625	112,125	7,475.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Vehicle type integration engineering (coordination with car builder, first-of-type test)	set	1	-	-	85,000.00	85,000	85,000	85,000.00	EUR	-	Priced for 15 vehicles; per-vehicle unit price in summary.
Per-vehicle installation + static/dynamic testing (with lineside loop/SKP)	adet	15	-	-	9,500.00	142,500	142,500	9,500.00	EUR	-	Priced for 15 vehicles; per-vehicle unit price in summary.
			3001.C Subtotal	567,000		281,725	848,725				

3001.D - Signalling Central Equipment & System Software (Seq. No 232)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Signalling main processor (server) – hot standby, with data storage	adet	2	48,000.00	96,000	4,800.00	9,600	105,600	52,800.00	EUR	TBD (to be confirmed)	
Server console: 21in LCD monitor + keyboard + mouse (KVM)	takim	1	3,800.00	3,800	380.00	380	4,180	4,180.00	EUR	TBD (to be confirmed)	
19in signalling equipment cabinet (TCC equipment room) + PDU	adet	1	9,500.00	9,500	1,425.00	1,425	10,925	10,925.00	EUR	TBD (to be confirmed)	
Main line signalling operator workstation + 2x 21–24in LCD monitors + keyboard + mouse	takim	2	12,500.00	25,000	1,250.00	2,500	27,500	13,750.00	EUR	TBD (to be confirmed)	
Depot signalling operator workstation + 21in LCD + keyboard + mouse	takim	1	8,500.00	8,500	850.00	850	9,350	9,350.00	EUR	TBD (to be confirmed)	
Signalling LAN ethernet switch (redundant) + patch/CAT6 + IP/serial converter	takim	2	7,500.00	15,000	1,875.00	3,750	18,750	9,375.00	EUR	TBD (to be confirmed)	
Signalling application software (main application, graphic interface/mimic, communication protocols, SKP reset / depot point command) + licences	set	1	260,000.00	260,000	39,000.00	39,000	299,000	299,000.00	EUR	TBD (to be confirmed)	
Vehicle and signalling system simulator software + simulator PC	set	1	95,000.00	95,000	-	-	95,000	95,000.00	EUR	TBD (to be confirmed)	
Interfaces: AVLS (vehicle/route/driver number), SCADA, NTP time synchronisation, PIS	set	1	85,000.00	85,000	-	-	85,000	85,000.00	EUR	TBD (to be confirmed)	
System engineering services (system integration, interface management)	ay	12	-	-	9,500.00	114,000	114,000	9,500.00	EUR	-	
Training (10 trainees, min 2 weeks / 10 working days) + training documentation	set	1	-	-	45,000.00	45,000	45,000	45,000.00	EUR	-	

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
Maintenance / fault-finding software + laptop diagnostic computer	set	1	28,000.00	28,000	2,800.00	2,800	30,800	30,800.00	EUR	TBD (to be confirmed)	
Integration with existing depot control centre – adaptation and licence	set	1	75,000.00	75,000	7,500.00	7,500	82,500	82,500.00	EUR	TBD (to be confirmed)	
			3001.D Subtotal	700,800		226,805	927,605				

3002 - Vehicle Tracking System (AVLS) - On-Board (Seq. No 233)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
On-board AVLS computer (EN 50155, min 48-channel GPS receiver, GSM/LTE modem, odometry input)	adet	15	7,200.00	108,000	-	-	108,000	7,200.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Integrated GSM/GPS roof antenna + antenna cable	adet	15	1,250.00	18,750	750.00	11,250	30,000	2,000.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Driver HMI terminal (delay/run messages, driver and run number entry) – 1 per cab	adet	30	2,400.00	72,000	360.00	10,800	82,800	2,760.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
On-board interface: odometry/speed, vehicle PIS/automatic announcement, signalling panel (vehicle/run number), wiring and installation kit	set	15	3,600.00	54,000	540.00	8,100	62,100	4,140.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
M2M data SIM / APN activation (installation and test period)	adet	15	180.00	2,700	27.00	405	3,105	207.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Depot wireless access point for data download/configuration (depot parking and workshop)	adet	2	2,800.00	5,600	1,680.00	3,360	8,960	4,480.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Lineside positioning support: transfer of signalling loop position data to AVLS (no additional field equipment)	set	1	18,000.00	18,000	3,600.00	3,600	21,600	21,600.00	EUR	TBD (to be confirmed)	Priced for 15 vehicles; per-vehicle unit price in summary.
Installation, per-vehicle testing and commissioning – AVLS field/on-board	set	1	-	-	65,000.00	65,000	65,000	65,000.00	EUR	-	
			3002 Subtotal	279,050		102,515	381,565				

3003 - Tram Vehicle Tracking System - Central Equipment and Software (Seq. No 234)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
AVLS server (hot standby) – TCC equipment room	adet	2	42,000.00	84,000	4,200.00	8,400	92,400	46,200.00	EUR	TBD (to be confirmed)	
Data recording / archive storage unit (NAS/SAN)	adet	1	18,000.00	18,000	2,700.00	2,700	20,700	20,700.00	EUR	TBD (to be confirmed)	
AVLS operator workstation + 21–24in monitor + keyboard + mouse	takim	1	9,000.00	9,000	900.00	900	9,900	9,900.00	EUR	TBD (to be confirmed)	
AVLS wall-mounted monitor 62–65in (professional 7/24) + wall/floor mounting kit	adet	1	7,500.00	7,500	750.00	750	8,250	8,250.00	EUR	TBD (to be confirmed)	
GSM/APN VPN router + firewall (redundant)	adet	2	6,500.00	13,000	650.00	1,300	14,300	7,150.00	EUR	TBD (to be confirmed)	
AVLS LAN ethernet switch + patch/CAT6	adet	1	4,200.00	4,200	1,050.00	1,050	5,250	5,250.00	EUR	TBD (to be confirmed)	
AVLS central software + licences (linear line map, overspeed, delay, reporting, arrival time calculation, PIS and signalling loop interface)	set	1	185,000.00	185,000	37,000.00	37,000	222,000	222,000.00	EUR	TBD (to be confirmed)	
Integration with the Municipality's existing AVLS system (software revision + physical/logical data link)	set	1	65,000.00	65,000	6,500.00	6,500	71,500	71,500.00	EUR	TBD (to be confirmed)	
Installation inside 19in cabinet (share of signalling/telecom cabinet) + wiring	set	1	6,500.00	6,500	3,900.00	3,900	10,400	10,400.00	EUR	TBD (to be confirmed)	
Central installation, integrated testing, training	set	1	-	-	55,000.00	55,000	55,000	55,000.00	EUR	-	

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
			3003 Subtotal	392,200		117,500	509,700				

3004 - Vehicle Tracking System Design (Seq. No 235)

Item	Unit	Qty	Material Unit EUR	Material Total EUR	Labour Unit EUR	Labour Total EUR	Total EUR	Installed Unit EUR	Cur.	Make / Model	Company Note (Firma Notu)
System architecture, communication infrastructure, data transmission structure, server and network topology, redundancy scenarios	set	1	-	-	25,000.00	25,000	25,000	25,000.00	EUR	-	
Vehicle positioning design (GPS + odometry + signalling loop fusion), on-board equipment layout and coordination with car builder (15 vehicles)	set	1	-	-	35,000.00	35,000	35,000	35,000.00	EUR	-	
Control centre screens, operator interfaces, alarm/reporting scenarios, data recording structures	set	1	-	-	22,000.00	22,000	22,000	22,000.00	EUR	-	
Cyber security requirements (GSM/APN, link to Municipality's existing AVLS) and integration principles	set	1	-	-	28,000.00	28,000	28,000	28,000.00	EUR	-	
Detail design, single-line/architecture diagrams, layout drawings, technical calculations and reports, test scenarios, revisions	set	1	-	-	40,000.00	40,000	40,000	40,000.00	EUR	-	
			3004 Subtotal	0		150,000	150,000				

SCOPE GAPS, DEVIATIONS AND POINTS FOR YOUR REVIEW

Subject	Our position	Action requested
Station Passenger Information (PIS)	Specification Section 104 para. 4.12 lists the station passenger information subsystem (central computer, software licences, platform LED displays) within the scope of this works, but the BOQ contains no corresponding line. Not included in this offer.	Please confirm whether PIS is to be included. If so, we will price it separately once the number of platforms and display types is confirmed.
Two-year spare parts	Required by Specification Section 103 para. 3.11; no corresponding BOQ line.	Not included. Please advise the required scope and we will price it separately.
Track circuits vs axle counters	BOQ describes 'ray devresi / kutle-agirlik dedektoru'. We have priced CASCO axle counters (EUR 4,450/unit) as the train detection solution.	Please confirm that axle counters are acceptable in lieu of track circuits.
Independent Safety Assessment (ISA)	Specification Section 103 para. 3.13 requires assessment by an engineer-approved independent body. Provisionally EUR 180,000.	Please advise whether the ISA body is nominated by the Engineer, and whether the cost is to be carried by us.
TSKP road traffic controllers and heads	6 controllers, 20x type-300 and 44x type-200 LED heads, 64 poles - road traffic equipment outside the core signalling supply.	Please confirm the local existing type and power interface so we can finalise these rates.
Line length	Your e-mail states 6.56 km; BOQ chainage KM 7+700 - 14+144.764 gives 6.444 km.	Priced on the BOQ chainage. Please confirm if the 6.56 km figure should apply.
Inductive loop and interrogator equipment	Loops, interrogators and loop cards have no equivalent in our previous CASCO supplies and are priced as engineering estimates.	To be confirmed with CASCO. Rates may be revised once the product range is confirmed.
Make / model list	Brands are confirmed for axle counters, the SIL3/SIL2 platform and tram-type point machines. Specific model numbers are still being confirmed with CASCO.	Full make/model list and technical data sheets will be issued before contract signature.
Installation interface	Our labour scope covers on-site installation, testing, commissioning and 15-day trial operation.	Excludes civil works, excavation, road reinstatement, UPS and LV cable trays (electrical package). Interface with the main contractor to be agreed.
Point machine quantities	Main line 16 (12 motorised + 4 manual), depot 40 (20 motorised + 20 manual), total 56.	Checked against the e-mail and BOQ note 3 - no discrepancy.

Quantities are not binding. As stated in your e-mail, quantities were taken off the tender drawings by your office. Where we consider a quantity or an item to be uncertain, we have noted it in the "Company Note (Firma Notu)" column of each line. Where our proposed solution differs from the BOQ description, this is flagged above for your confirmation before award.

BASIS OF PRICING - COMPARISON WITH PREVIOUS CASCO QUOTATIONS

Item	CASCO Konya (10.06.2026)	CASCO Adapazari (13.06.2026)	This offer	Note
Interlocking / SIL control unit	165,000	165,000	145,000 / 110,000	SIL3 SKP (main line) and SIL2 SKP (depot), stepped down from SIL4
OCC / control centre	220,000	220,000	Itemised	Workstations, servers and software priced line by line
Tram-type point machine (IP67)	4,864 (S600)	6,625	6,625 / 5,900	Main line from Adapazari; depot slightly reduced
Axle counter / train detection	4,450	4,650	4,450	Konya rate; axle counters substituted for track circuits

Item	CASCO Konya (10.06.2026)	CASCO Adapazari (13.06.2026)	This offer	Note
Signal head	1,290	1,290	1,450 - 1,850	Uplifted for number of aspects
Site services (testing & commissioning)	1,100,000	excluded	2,518,959	Split into design / commissioning / training / system engineering

Unit rates are carried over from the two previous CASCO quotations (both turnkey tram SIL4 schemes, FOB, EUR excl. VAT) and adjusted for the SIL3/SIL2 levels and scope breakdown of this project. Those quotations excluded design and ISA; both have been estimated separately here. Lines with no historical equivalent (loops, interrogators, AVLS, on-board equipment, software) are engineering estimates.

COMMERCIAL TERMS

- Prices are DAP Sanliurfa site, EUR, VAT excluded. Currency EUR is stated on every line.
- Validity: 180 calendar days from 1 October 2026.
- Delivery: design within 3 months of contract effectiveness; equipment in batches over 6-9 months, batching to be agreed against project progress.
- Standards: EN 50126, EN 50128, EN 50129, EN 50121-4, EN 61508, EN 50155, ETSI EN 300 330, and compliance with the Turkish KET regulation.
- Warranty: 24 months from completion of commissioning or 30 months from delivery, whichever comes first.
- FAT: the purchaser may attend routine factory acceptance tests.
- Excluded: civil works, excavation, road reinstatement, UPS, LV power cable trays, import customs duty and import taxes.
- Technical data sheets and the full make/model list will be issued before contract signature.

Sources: BOQ URF_1Etap_SINYALIZASYON_Teklif_Talebi.xlsx (29.09.2026); Specification Sections 103 and 104; enquiry e-mail from Serhat CELIK (MEP Center), 1 October 2026, 21:07; previous CASCO quotations Konya (EUR 4,440,670) and Adapazari (EUR 3,364,100). **FX:** EUR/CNY 7.5692 / EUR/TRY 55.3528 - Sina Finance, 1 October 2026, 21:49 GMT+8. **Legend:** "model TBC" = brand confirmed, model number to be confirmed with CASCO; "TBD" = to be determined; "Ongoru" = quantity was an estimate in the BOQ.